COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	7/9/2026	**DATE SUBMITTED:**	7/30/2026
NAME:	Momo, Borge Rudolf A.	**INSTRUCTOR:**	Engr. Quejado

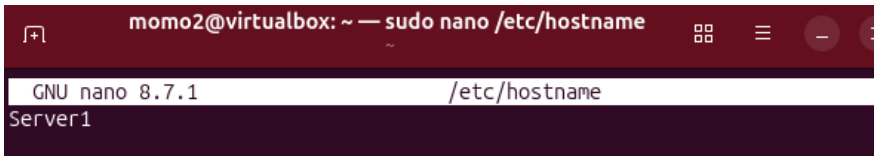
ACTIVITY NO. 1.1 Creating Virtual Machines

1. PROCEDURE OUTPUT

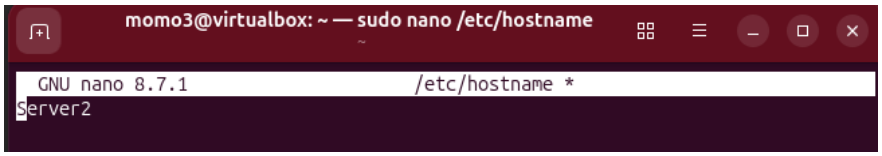
Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname

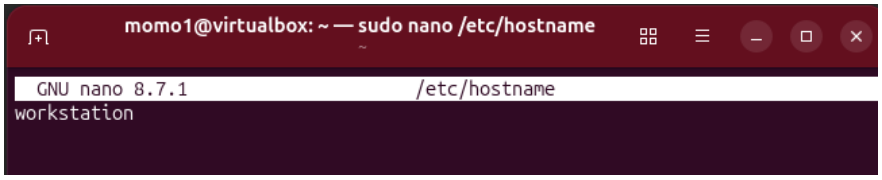
1.1 Use server1 for Server 1



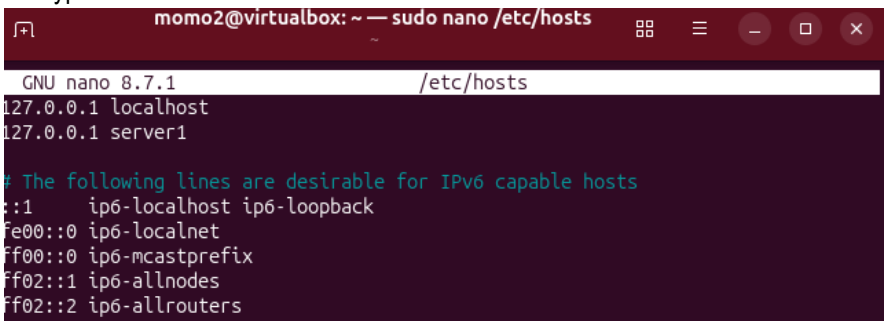
1.2 Use server2 for Server 2



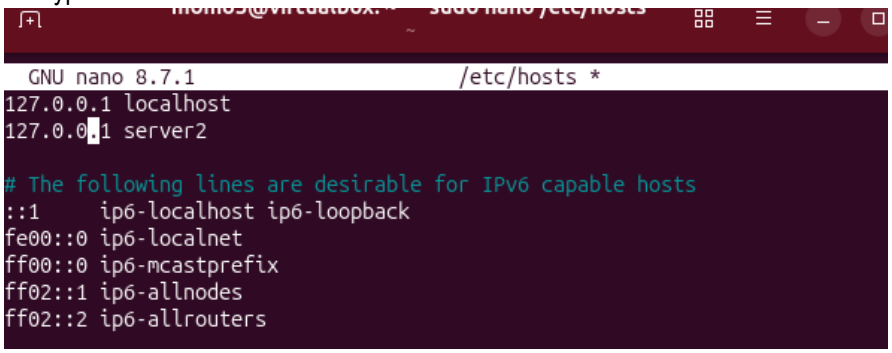
1.3 Use workstation for the Local Machine



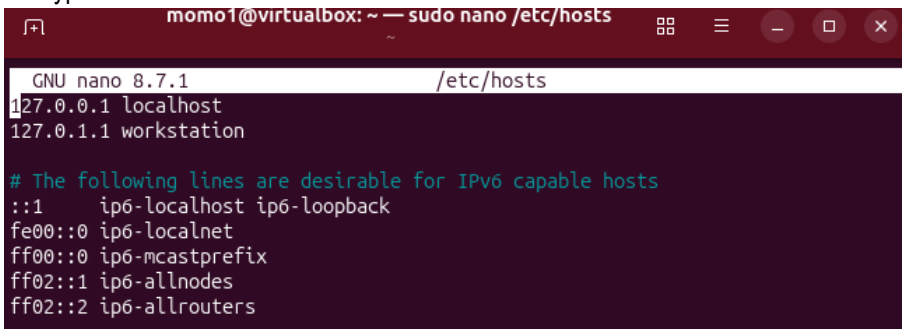
2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



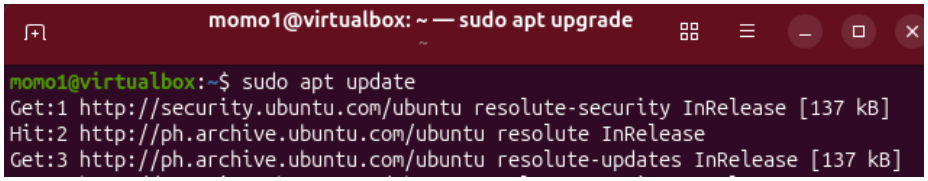
2.3 Type 127.0.0.1 workstation for the Local Machine

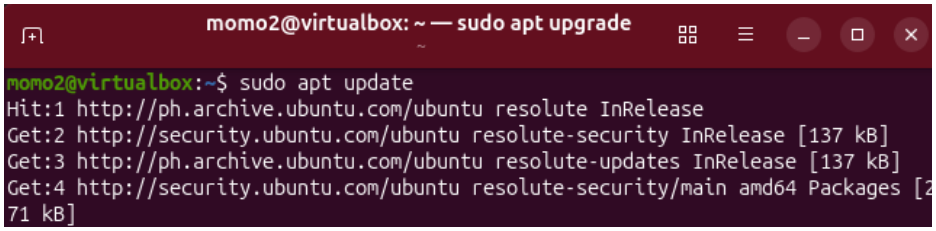


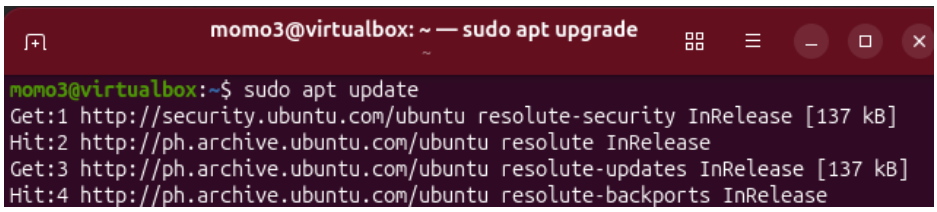
Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.

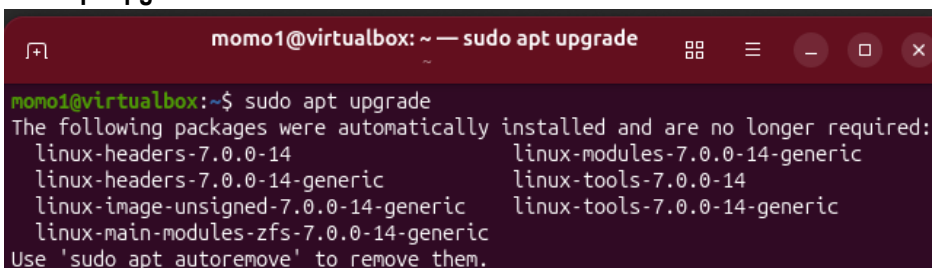
Sudo apt update

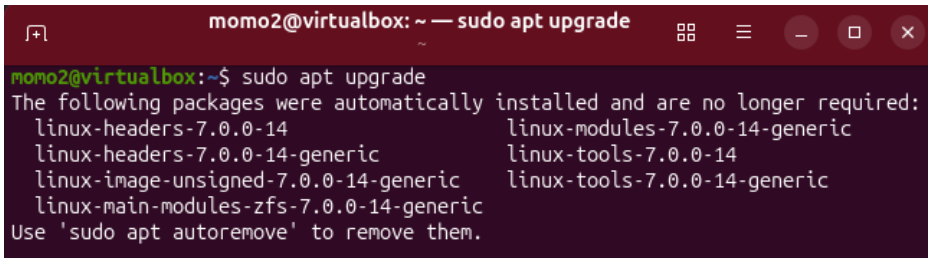


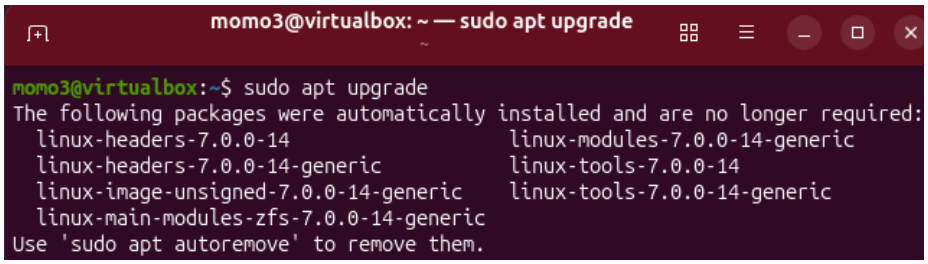




Sudo apt upgrade







2. Install the SSH server using the command sudo apt install openssh-server.

workstation

```
momo1@virtualbox:~$ sudo apt install openssh-server
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                  linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic          linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic   linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  openssh-server
```

server1

```
momo2@virtualbox:~$ sudo apt install openssh-server
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                  linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic          linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic   linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  openssh-server
```

server2

```
momo3@virtualbox:~$ sudo apt install openssh-server
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                  linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic          linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic   linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  openssh-server
```

3. Verify if the SSH service has started by issuing the following commands:

3.1 sudo service ssh start

```
momo1@virtualbox:~$ sudo service ssh start
momo2@virtualbox:~$ sudo service ssh start
momo3@virtualbox:~$ sudo service ssh start
```

3.2 sudo systemctl status ssh

```
momo1@virtualbox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena
     Active: active (running) since Thu 2026-07-09 01:58:16 UTC; 1min 55s ago
 Invocation: 1224d45bfbd14ebf957d5ec53aaca42a
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
```

```
momo2@virtualbox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena
     Active: active (running) since Thu 2026-07-09 01:58:37 UTC; 2min 24s ago
 Invocation: fa2ba210a8a64595b335ef706bcd26d7
TriggeredBy: ● ssh.socket
```

```
momo3@virtualbox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena
     Active: active (running) since Thu 2026-07-09 01:58:45 UTC; 14s ago
 Invocation: 72f4f4dde2aa4db5aee13ec1a3c50a65
TriggeredBy: ● ssh.socket
```

4. Configure the firewall to all port 22 by issuing the following commands:
4.1 sudo ufw allow ssh
4.2 sudo ufw enable
4.3 sudo ufw status

workstation

```
momo1@virtualbox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
momo1@virtualbox:~$ sudo ufw enable
Firewall is active and enabled on system startup
momo1@virtualbox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

server1

```
momo2@virtualbox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
momo2@virtualbox:~$ sudo ufw enable
suFirewall is active and enabled on system startup
momo2@virtualbox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

server2

```
momo3@virtualbox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
momo3@virtualbox:~$ sudo ufw enable
Firewall is active and enabled on system startup
momo3@virtualbox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:
1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all The machines are in this network 192.168.56.XX.
1.1 Local Machine (workstation) IP address: 192.168.56.**113**

```
192.168.56.113
```

1.2 Server 1 IP address: 192.168.56.**115**

```
inet 192.168.56.115
```

1.3 Server 2 IP address: 192.168.56.**114**

```
inet 192.168.56.114
```

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☑ Successful ☐ Not Successful

```
momo@workstation:~$ ping 192.168.56.115
PING 192.168.56.115 (192.168.56.115) 56(84) bytes of data.
64 bytes from 192.168.56.115: icmp_seq=1 ttl=64 time=0.832 ms
64 bytes from 192.168.56.115: icmp_seq=2 ttl=64 time=1.32 ms
64 bytes from 192.168.56.115: icmp_seq=3 ttl=64 time=1.46 ms
64 bytes from 192.168.56.115: icmp_seq=4 ttl=64 time=1.33 ms
^C
--- 192.168.56.115 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3054ms
rtt min/avg/max/mdev = 0.832/1.234/1.457/0.238 ms
```

2.2 Connectivity test for Local Machine 1 to Server 2: ☑ Successful ☐ Not Successful

```
momo@workstation:~$ ping 192.168.56.114
PING 192.168.56.114 (192.168.56.114) 56(84) bytes of data.
64 bytes from 192.168.56.114: icmp_seq=1 ttl=64 time=4.10 ms
64 bytes from 192.168.56.114: icmp_seq=2 ttl=64 time=0.439 ms
64 bytes from 192.168.56.114: icmp_seq=3 ttl=64 time=0.429 ms
64 bytes from 192.168.56.114: icmp_seq=4 ttl=64 time=0.980 ms
64 bytes from 192.168.56.114: icmp_seq=5 ttl=64 time=0.608 ms
^C
--- 192.168.56.114 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 5526ms
rtt min/avg/max/mdev = 0.429/1.310/4.095/1.406 ms
```

2.3 Connectivity test for Server 1 to Server 2: ☑ Successful ☐ Not Successful

```
momo@server1:~$ ping 192.168.56.114
PING 192.168.56.114 (192.168.56.114) 56(84) bytes of data.
64 bytes from 192.168.56.114: icmp_seq=1 ttl=64 time=1.46 ms
64 bytes from 192.168.56.114: icmp_seq=2 ttl=64 time=0.661 ms
64 bytes from 192.168.56.114: icmp_seq=3 ttl=64 time=0.371 ms
64 bytes from 192.168.56.114: icmp_seq=4 ttl=64 time=1.32 ms
^C
--- 192.168.56.114 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3082ms
rtt min/avg/max/mdev = 0.371/0.953/1.460/0.452 ms
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

1.2 Enter the password for server 1 when prompted

```
momo@workstation:~$ ssh momo@192.168.56.115
The authenticity of host '192.168.56.115 (192.168.56.115)' can't be established.
ED25519 key fingerprint is SHA256:FwwF11OhlUOIdn8ES3Xn8kXZmqtX2Dy7/vnhfNL23Xk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.115' (ED25519) to the list of known hosts
.
momo@192.168.56.115's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

1.3 Verify that you are on server 1. The user should be in this format user@server1.
For example, jvtaylar@server1

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

momo@server1:~$
```

2. Logout of Server 1 by issuing the command control + D.

3. Do the same for Server 2.

```
momo@server1:~$ ssh momo@192.168.56.114
The authenticity of host '192.168.56.114 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:FwwF11OhlUOIdn8ES3Xn8kXZmqtX2Dy7/vnhfNL23Xk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.114' (ED25519) to the list of known hosts
.
momo@192.168.56.114's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

Verifying:

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

momo@server2:~$
```

4. Edit the hosts of the Local Machine by issuing the command sudo nano /etc/hosts. Below all texts type the following:
4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

```
  GNU nano 7.2                          /etc/hosts *
127.0.0.1 localhost
127.0.0.1 workstation
192.168.56.115 server1
192.168.56.114 server2
# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

4.3 Save the file and exit.
5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Connecting to Server 1:

```
momo@workstation:~$ ssh momo@server1
The authenticity of host 'server1 (192.168.56.115)' can't be established.
ED25519 key fingerprint is SHA256:FwwF11OhlUOIdn8ES3Xn8kXZmqtX2Dy7/vnhfNL23Xk.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
momo@server1's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 16 00:44:26 2026 from 192.168.56.113
momo@server1:~$
```

Connecting to Server2

```
momo@workstation:~$ ssh momo@server2
The authenticity of host 'server2 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:FwwF11OhlUOIdn8ES3Xn8kXZmqtX2Dy7/vnhfNL23Xk.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
momo@server2's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 16 00:41:41 2026 from 192.168.56.115
momo@server2:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?

- We are able to use the hostname instead of IP address in a way that the computer uses DNS to find the correct IP Address. This makes it easier to remember and connect to another computer and it is more convenient than typing a long IP address every time.

2. How secured is SSH?

- SSH is very much secure since it encrypts the data that was sent between computers. This helps protect passwords and other information from hackers. It is widely used for safe remote access to computers.

CONCLUSIONS

In conclusion, I learned how to create and configure virtual machines for server management. I also learned how to set up SSH and test the connection between different machines. Overall, it improved my learnings in using Linux commands and managing virtual machines and I hope to do more of these in the future since it's simple yet fun.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: **Formatting**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The AI tool was only used for the purpose of formatting my sentences for supplementary activity.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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